Project Ariel: Guided Divergence and Grounded Return — Art-Based Canvas Storytelling with AI Threat-Appraisal Gating for Students

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Status. Conceptual framework, not validated science. This paper proposes an educational design, a formal loop, and a toy simulation. It is not a diagnostic instrument, not a clinical protocol, and not a substitute for a teacher, counselor, parent, or licensed professional. In the U.S., call or text 988 for crisis support; for immediate danger or a medical emergency, call 911. Involve a trusted adult when a student is at risk.

Abstract

Students need room for abstract, unguided, divergent thinking—and a reliable route back to the shared world of evidence, skills, and next steps. Building on the Project Ariel gated-line model, we propose *Guided Divergence and Grounded Return*: learners explore freely through canvas art and narrative storytelling; an AI monitors noisy lexical proxies of threat appraisal (A); if a story tips into unconstructive rumination and the gate narrows, the system issues graded Socratic grounding challenges until A falls below a hysteretic threshold; it then identifies the learner's intent and offers a factual knowledge path that bridges grandiose goals to the first concrete step. A deterministic session simulation shows threat rising under a ruminative narrative ($A_{\text{peak}} = 0.604$), a challenge window ($t \approx 19.4$–44.4), gate recovery, and knowledge-path progress reaching 1.0 as distance-to-reality falls to 0.06. We map the loop onto the live Ariel Suite UI, propose Challenge / Intent / Knowledge-Path panels, and outline an ethics-bound classroom pilot. Lexical threat estimates are proxies, not diagnoses; imagination is not pathology.

1 Introduction: Leave—and return

Education that only drills convergent answers can starve curiosity. Education that only celebrates fantasy can strand a student in unfinishable grandeur. The design question is narrower: **How can a learning environment invite open, abstract, unguided exploration while still offering a respectful, evidence-seeking path back to reality?**

Project Ariel previously formalized state-dependent meaning transfer: a retained value or goal can coexist with a narrowed route to action when threat rises and regulation falls. This paper re-aims that machinery at students. The canvas and journal become instruments of *guided divergence*; the gated engine becomes a *return mechanism* that does not shame imagination, but challenges rumination and then scaffolds a factual ladder from a grand goal to a first step.

2 Theoretical foundations

Divergent and convergent thinking. Guilford's mid-century call for creativity research distinguished production of many alternatives from single-answer convergence [1]. His

later Structure-of-Intellect work elaborated divergent production as a family of abilities [2]. Ariel treats the canvas as a divergence space and the knowledge path as a convergence scaffold—not as a hierarchy of worth.

Repetitive thought is not all one thing. Watkins reviews constructive and unconstructive repetitive thought, emphasizing valence, context, and abstract versus concrete construal [3]. Depressive rumination—passive, repetitive focus on distress and its implications—is associated with maintained depression and impaired problem solving [4]. Ariel therefore tries to protect exploratory narrative while detecting when a story becomes stuck, self-attacking, or hopeless in tone.

From free fantasy to binding goals. Oettingen and colleagues show that freely fantasizing a desired future without confronting obstacles can leave commitment weak; mental contrasting of future and reality can create expectancy-dependent goals [5]. Implementation intentions—"if situation Y, then I will do Z"—further support enactment [6]. The Ground and Direct phases of Ariel echo Wish/Outcome/Obstacle/Plan (WOOP/MCII) without claiming to deliver a clinical protocol.

Guided discovery, not argument. In CBT practice, Socratic questioning aims at guided discovery: questions the learner can answer, attention to overlooked evidence, and movement from concrete to abstract so the learner rebuilds a conclusion [7]. Ariel's Challenge panel is modeled on that spirit—graded prompts about evidence, present sensory detail, and what one would tell a friend—not on lecturing a student out of a dream.

Expression, autonomy, and scaffolding. Expressive writing about emotional experience has been linked, under boundary conditions, to health-relevant outcomes [8]; art-making is treated here cautiously as a related expressive medium, not as proven art therapy. Self-determination theory highlights autonomy, competence, and relatedness as supports for engagement [9]. Scaffolding temporarily reduces degrees of freedom while preserving the learner's agency [10]. Vygotsky's zone of proximal development supplies the educational metaphor: the AI may assist just beyond independent reach, then withdraw.

Lexical proxies are noisy. Absolutist words (e.g., *always*, *never*, *completely*) appear elevated in some online language samples associated with anxiety, depression, and suicidal ideation forums [11]. LIWC-style word-category methods, used widely in expressive-writing research, quantify attention and emotion in text but do not read minds [8]. Ariel treats such features as *hypothesis generators* for threat appraisal, never as diagnoses.

3 The Guided Divergence loop

Five phases form a cycle that can repeat within a session:

1. **Explore.** Canvas art and storytelling with the gate relatively open. Divergent prompts are allowed; no premature convergence.
2. **Appraise.** An AI estimates threat A from narrative and (optionally) canvas features: absolutist density, self-focus, repetitive theme recurrence, hopelessness markers. All are noisy.

3. **Challenge (conditional).** If A exceeds an upper threshold and a rumination loop is suspected, issue graded Socratic grounding prompts until A falls below a lower threshold (hysteresis).
4. **Ground.** Identify intent behind the fantasy (mastery, recognition, belonging, escape, contribution, justice, play, ...).
5. **Direct.** Offer a factual knowledge path: prerequisites, milestones, first concrete step, obstacle contrast, and an if–then plan.

3.1 Threat estimate

Let $f_{\text{abs}}, f_{\text{rep}}, f_{\text{self}}, f_{\text{hope}} \in [0,1]$ be normalized feature intensities from text (and optionally stroke entropy on the canvas). A simple proxy is

$$\hat{A}_t = \text{clip}_{[0,1]}\Big(a_0 + w_1 f_{\text{abs}} + w_2 f_{\text{rep}} + w_3 f_{\text{self}} + w_4 f_{\text{hope}}\Big), \tag{1}$$

with illustrative weights $(a_0, w_1, w_2, w_3, w_4) = (0.12, 0.35, 0.30, 0.20, 0.25)$. The displayed threat state relaxes toward the estimate:

$$\tau_A \dot{A} = \hat{A}_t - A. \tag{2}$$

During Challenge, prompts pull the target downward (graded support), so $\hat{A}$ is mixed with a lower set-point. These coefficients are design choices, not fitted clinical parameters.

3.2 Hysteretic gate and suite alignment

Available control r rises when grounding succeeds and falls when threat dominates. The *paper* form of the gated line was $g = 0.05 + 0.95\,P(1-G)$. The **live Ariel Suite** implements a UI simplification using threat v and control r [14]:

$$g = \text{clamp}\big(0.05 + 0.95\,(0.2 + 0.8\,r)\,(1 - 0.3\,v),\ 0.05,\ 1\big). \tag{3}$$

In this student model we set $v \approx A$. Challenge mode engages when $A \geq \theta_{\text{hi}}$ and releases only when $A \leq \theta_{\text{lo}}$ with $\theta_{\text{hi}} > \theta_{\text{lo}}$ (hysteresis), preventing flicker.

3.3 Challenge intensity

Let $c_t \in [0,1]$ be challenge intensity. While Challenge is active, c rises; otherwise it decays:

$$\dot{c} = \begin{cases} \kappa_+\,(1-c) & \text{in Challenge,} \\ -\kappa_-\,c & \text{otherwise.} \end{cases} \tag{4}$$

Prompt difficulty is scheduled by c: sensory/present-moment first; then concrete evidence; then friend-advice and next observable step. The AI must not override the student's values; it may only request specificity.

3.4 Distance-to-reality and knowledge progress

Let $G^\star$ be a grandiose goal embedding (or a structured skill graph node) and P_k the current knowledge-path node. Define a distance-to-reality

$$d_t = \delta(G^\star, P_k) \in [0,1], \tag{5}$$

high when the fantasy is far from any first concrete step. Knowledge-path progress p_t increases when the gate is open enough and Challenge has cooled:

$$\dot{p} = \eta\, g\,(1-A)\,\mathbf{1}_{\{\text{Ground/Direct}\}}, \qquad \dot{d} = -\zeta\, g\,(1-A)\,\mathbf{1}_{\{\text{Ground/Direct}\}}. \tag{6}$$

Progress is educational scaffolding, not a grade and not a moral score.

4 Simulation of one session

We integrate the loop with forward Euler ($\Delta t = 0.05$) over $T = 80$ arbitrary time units. A synthetic narrative feature schedule rises through a ruminative middle segment, then falls under Challenge. Thresholds: $\theta_{\text{hi}} = 0.52$, $\theta_{\text{lo}} = 0.34$. Halving Δt changes the final state by $< 10^{-6}$.

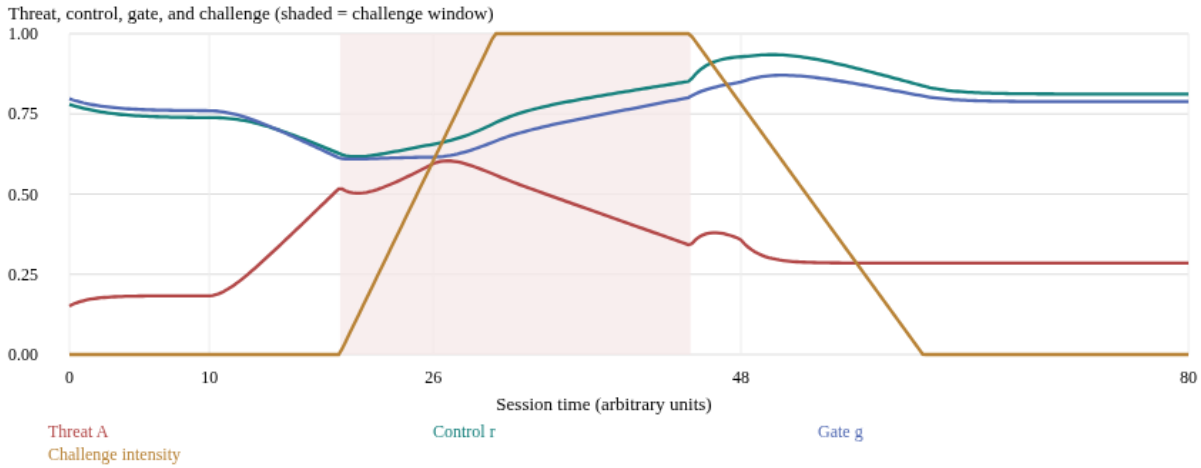


Figure 1: Threat A, control r, suite-aligned gate g, and challenge intensity. Shaded band marks the Challenge window ($t \approx 19.4$–44.4).

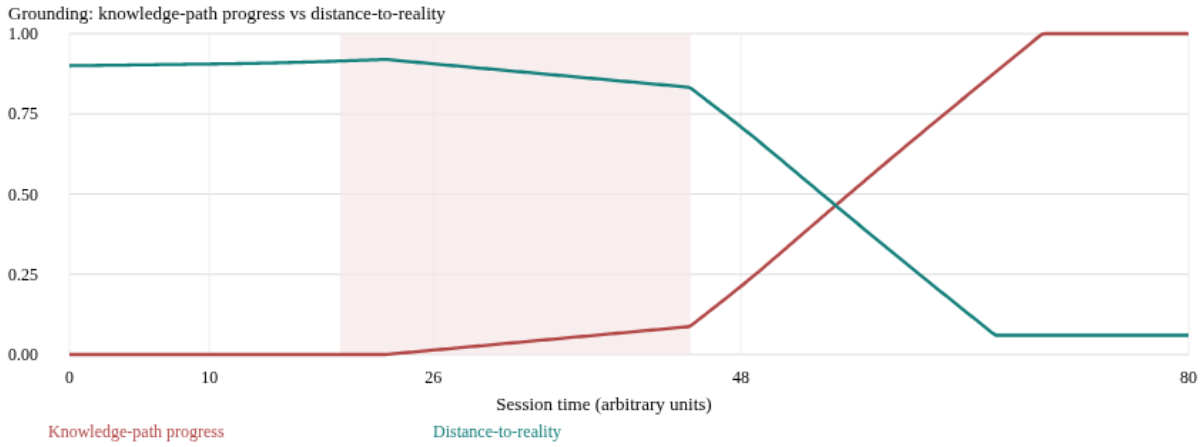


Figure 2: Knowledge-path progress rises after grounding; distance-to-reality falls. Values are synthetic, not student data.

Reproduce with `python3 students/simulate_session.py`. The run illustrates the designed loop; it does not prove learning gains.

5 Worked example: “Design a city on Mars”

Explore. A student paints red domes and writes a story of being the architect who finally “gets it right when Earth never listened.”

Table 1: Key synthetic session numbers.

Quantity	Value
A peak / time	0.604 / $t = 27$
Challenge enter / exit	$t = 19.35$ / $t = 44.4$
A at Challenge exit	0.346
g at peak A / final g	0.619 / 0.788
Final progress / distance	1.00 / 0.06

Appraise. Absolutist and hopeless markers rise ("never," "always alone," "completely impossible unless I..."); A crosses θ_{hi}.

Challenge. Graded prompts: What color is the desk in front of you? What is one sentence a friend would say? What is one fact about Mars you could check in a reliable source today?

Ground (intent). Classification (illustrative, editable by the student): primary intent = *mastery* + *contribution*; secondary = recognition; escape is present but not endorsed as the goal.

Direct (factual path). An ordered ladder—labeled **illustrative**, not a career prescription:

1. Pre-algebra fluency $\rightarrow$ Algebra I (equations, graphs).
2. Conceptual physics (forces, energy) with lab notebooks.
3. Chemistry foundations (matter, reactions) for life-support literacy.
4. Intro engineering / design thinking: constraints, trade-offs, prototypes.
5. A small CAD or physical model of *one* habitat module (not a city).
6. Join a mentor, club, or maker space; write one if–then study plan for the next week.

Mental contrasting names a real obstacle (e.g., "I lose Fridays to scrolling"); an implementation intention pairs it with a cue ("If it is Friday 4pm, then I open the algebra set for 25 minutes").

6 Mapping to the live Ariel Suite

The public suite at `https://ryryespi.github.io/mindframe/ariel/suite/` already joins four panes: Drawing Canvas, Journal Mapping, Gate engine, and Charts [14]. Bridge code exposes

$$g = \text{clamp}(0.05 + 0.95(0.2 + 0.8r)(1 - 0.3v), 0.05, 1)$$

and a transfer readout along I$\rightarrow$O. Proposed additions for Guided Divergence:

- **Challenge panel:** timed, graded Socratic cards; student can skip, edit, or call a human.
- **Intent card:** multi-label intent chips the student confirms or rejects.
- **Knowledge Path ladder:** ordered milestones with "first concrete step" highlighted; exports to a teacher view.

Journal nodes (Retention, Transfer, Execution, Support, Grace) remain available; for classrooms, Grace is optional and never coerced.

7 Ethics and safety

Imagination is not a disorder. Challenge must not pathologize play, grief, culture, or neurodiversity. Teachers or counselors remain in the loop for minors; obtain consent/assent; prefer local-first storage; minimize data leaving the device. Lexical proxies inherit cultural and linguistic bias—absolutist words in one dialect are not pathology. The AI must not override student values or religious commitments. If self-harm, hopelessness about living, or acute panic appears, stop the loop and escalate: trusted adult, school protocol, and in the U.S. 988 / 911 as appropriate. This system is not crisis care.

8 Proposed classroom pilot

Design. Feasibility pilot in 2–4 classrooms ($N \approx 48$ consenting students with guardian consent/assent as required), randomized by class period to Ariel Guided Divergence versus time-matched creative journaling without threat gating. Ethics review required. No provocation of distress; skip anytime.

Measures (age-appropriate forms). Alternative Uses Task (divergent production) [1, 2]; a brief repetitive-negative-thinking measure such as an age-appropriate RRS brooding subset or PTQ items where validated for the age band [4, 12]; goal-attainment scaling for the first concrete step [13]; engagement and autonomy-support ratings [9]. Primary: completion of one teacher-visible concrete step within two weeks. Secondary: change in repetitive-thought scores without loss of Alternative Uses fluency.

Predictions. (1) Divergence fluency need not fall when gating is present. (2) Students with rising lexical-A who receive Challenge will show more concrete next-step language than matched journaling alone. (3) A null result on step completion would challenge the Direct phase more than the value of art itself.

Limitations. Simulation $\neq$ classroom. Lexical A is biased and gameable. Grand goals are culturally shaped. Teachers need training so "challenge" does not become humiliation.

9 Conclusion

Guided Divergence and Grounded Return reframes Project Ariel for students: protect the freedom to think sideways, notice when a narrative closes the gate through rumination, challenge gently until threat proxies fall, name the intent beneath a fantasy, and lay a factual path toward a first real step. The live suite already carries canvas, journal, gate, and charts; Challenge, Intent, and Knowledge-Path panels are the natural next build. The honest claim today is a reproducible design hypothesis—not a finished pedagogy and not a clinical tool.

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All credit and glory to God. I owe my knowledge to my Lord Jesus Christ.